

US Candy Distributor Sales, Factory & Target Performance Dashboard

1. Project Overview

This project analyzes sales, profit, product, factory, regional, geographic, and target performance for a U.S. candy distributor. The goal was to transform raw distributor data into a clean analytical model using SQL Server and create an interactive Power BI dashboard that supports business decision-making.

The final dashboard provides a clear view of overall business performance, identifies the strongest products and factories, shows geographic sales concentration, and compares actual sales against division-level targets.

2. Business Objective

The objective of this project was to answer the following business questions:

- What are the company's total sales, gross profit, orders, and average order value?
 - Which candy divisions, products, and factories generate the most revenue and profit?
 - Which regions and states contribute the most to sales?
 - Where are customers geographically concentrated at ZIP-code level?
 - Which divisions are exceeding or missing their sales targets?
 - Are there any data quality issues that may affect business interpretation?
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3. Tools Used

- **SQL Server:** Data storage, cleaning, transformation, and reporting views
 - **SQL Server Management Studio:** Database creation, CSV import, SQL querying
 - **Power BI:** Dashboard design, visual analytics, KPI cards, charts, and maps
 - **Excel/CSV files:** Raw source data
 - **GitHub/Portfolio:** Project documentation and presentation
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4. Dataset Description

The dataset contains candy distributor sales and supporting lookup files. The main tables used were:

Table	Description
Candy Sales	Transaction-level sales, cost, profit, customer, shipping, product, and location data
Candy Products	Product details, product division, factory, unit price, and unit cost
Candy Factories	Factory names and geographic coordinates
Candy Targets	Division-level sales targets
US ZIP Codes	ZIP-level latitude, longitude, city, state, population, and location attributes

The raw sales table contained **10,194 rows** and included order-level details such as order date, ship mode, customer, product, sales, units, gross profit, and cost.

5. Data Preparation and SQL Workflow

A SQL Server database named **CandyDistributorDB** was created to store and analyze the raw data. The CSV files were imported into raw staging tables, then transformed into clean SQL views.

The main SQL workflow included:

1. Creating a dedicated project database.
2. Importing raw CSV files into SQL Server.
3. Reviewing table structures and column data types.
4. Checking row counts and missing values.

5. Standardizing ZIP codes for geographic joins.
6. Joining sales data with product, factory, target, and ZIP-code tables.
7. Creating clean views for reusable analysis.
8. Creating summary reporting views for Power BI.

Key SQL views created included:

- `vw_sales_clean`
 - `vw_products_clean`
 - `vw_factories_clean`
 - `vw_targets_clean`
 - `vw_zips_clean`
 - `vw_kpi_summary`
 - `vw_monthly_sales`
 - `vw_region_performance`
 - `vw_division_performance`
 - `vw_product_performance`
 - `vw_factory_performance`
 - `vw_target_performance`
 - `vw_state_performance`
 - `vw_geo_sales`
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6. Data Quality Findings

During data validation, a major issue was identified in the shipping date fields. The ship dates were far later than the order dates, causing unrealistic shipping durations across the dataset.

Because of this issue, shipping duration was not used as a key performance indicator. However, ship mode was still used for sales and order-volume analysis because the ship mode categories were still meaningful.

Additional validation showed:

- Product joins were complete.
- Factory joins were complete.
- Some ZIP codes did not match the ZIP-code reference table.
- ZIP-level mapping was still possible using matched ZIP records with valid latitude and longitude.

This data quality review was included in the dashboard to show transparency and analytical judgment.

7. Key Performance Indicators

The overall business performance was summarized using the following KPIs:

KPI	Value
Total Sales	\$141,783.63
Gross Profit	\$93,442.80
Total Cost	\$48,340.83
Units Sold	38,654
Total Orders	8,549
Total Customers	5,044
Profit Margin	65.91%
Average Order Value	\$16.58

These KPIs provide a high-level view of revenue, profitability, order volume, and customer activity.

8. Dashboard Pages

The Power BI dashboard was organized into five main pages.

Page 1: Executive Overview

The Executive Overview page provides a summary of overall business performance. It includes KPI cards for total sales, gross profit, total orders, average order value, and profit margin.

It also includes visuals for:

- Monthly sales trend
- Sales by region
- Sales by division
- Factory sales contribution
- Data quality note related to shipping dates

This page gives stakeholders a quick understanding of the company's overall performance.

Page 2: Product & Factory Performance

This page focuses on product and factory contributions. It identifies the products and factories that generate the most sales and profit.

Main visuals include:

- Top products by sales
- Top products by gross profit
- Factory sales contribution
- Factory profit margin
- Key business insight box

The analysis showed that chocolate products dominate both sales and gross profit. Lot's O' Nuts and Wicked Choccy's were the strongest factory contributors.

Page 3: Product Details

The Product Details page provides a table-level view of product performance. It includes product name, division, sales, gross profit, units sold, profit margin, and order count.

This page supports detailed product comparison and allows users to review exact values behind the summary visuals.

Page 4: Regional & Geographic Performance

This page analyzes sales performance by region, state, and ZIP-code location.

Main visuals include:

- Sales by region
- Top states by sales
- Top states by gross profit
- ZIP-level sales map
- Geographic insight summary

The analysis showed that the Pacific and Atlantic regions were the strongest sales regions. California and New York were leading states, and ZIP-level mapping showed sales activity spread across the United States with stronger concentration near major customer markets.

Page 5: Target Performance

This page compares actual sales against division-level sales targets.

Main visuals include:

- Actual sales KPI
- Target sales KPI
- Sales gap KPI
- Actual sales vs target by division
- Target achievement percentage by division
- Sales gap by division
- Target performance details table

The target analysis showed that Chocolate significantly exceeded its target and was the primary driver of overall target achievement. Others also exceeded its target, while Sugar substantially underperformed.

9. Major Business Insights

1. Chocolate drives the majority of sales and profit

Chocolate generated **\$131,692.90** in sales, making it the dominant division. It also produced the majority of gross profit, showing that it is the strongest business segment.

2. Sugar underperformed significantly

Sugar generated only **\$427.48** in actual sales compared to a target of **\$15,000**, reaching only **2.85%** of its target. This division may require a review of product strategy, pricing, distribution, or demand.

3. Lot's O' Nuts and Wicked Choccy's are the strongest factories

Factory analysis showed that Lot's O' Nuts and Wicked Choccy's contributed the highest sales and profit. These factories are important to the distributor's overall performance.

4. Pacific and Atlantic are the strongest sales regions

Regional analysis showed that Pacific and Atlantic were the top-performing regions. State-level analysis further showed that California and New York were major contributors.

5. Geographic mapping adds location-based insight

The ZIP-level map showed sales distribution across the United States. This helps identify where customer activity is concentrated and supports future regional sales or marketing planning.

6. Data quality review improved trust in the analysis

The shipping date issue was identified and clearly documented. Instead of using unreliable shipping duration metrics, the dashboard focused on trusted fields such as sales, profit, product, factory, region, and target performance.

10. Recommendations

Based on the analysis, the following recommendations are suggested:

1. **Prioritize Chocolate products** because they generate the highest sales and profit.
2. **Investigate Sugar division performance** to understand why it is far below target.
3. **Focus sales and distribution efforts in high-performing regions** such as Pacific and Atlantic.

4. **Use ZIP-level customer concentration to support regional marketing decisions.**
 5. **Monitor factory contribution**, especially for Lot's O' Nuts and Wicked Choccy's.
 6. **Improve source data quality**, especially shipping date accuracy, before using shipping duration as a performance metric.
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11. Skills Demonstrated

This project demonstrates the following data analytics skills:

- SQL database creation and data import
 - Data cleaning and transformation using SQL views
 - Data validation and quality checks
 - SQL joins across multiple business tables
 - KPI calculation
 - Aggregation and reporting view creation
 - Power BI dashboard design
 - Data visualization and business storytelling
 - Geographic mapping using latitude and longitude
 - Target performance analysis
 - Portfolio-ready project documentation
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12. Conclusion

The US Candy Distributor dashboard successfully transforms raw sales and operational data into a clear business intelligence solution. The project highlights key sales drivers, product and factory performance, regional sales concentration, and target achievement.

The analysis showed that Chocolate is the strongest division, Pacific and Atlantic are the leading regions, and Sugar requires further business review. The dashboard also documents important data quality limitations, making the final analysis more transparent and reliable.

This project demonstrates a complete end-to-end analytics workflow using SQL Server and Power BI, from raw data preparation to executive dashboard creation and business insight communication.